MIL-S-52872B
24 March 1980
SUPERSEDING
MIL-S-52872A(ME)
7 March 1977

MILITARY SPECIFICATION

SENSOR GROUP, ULTRASONIC MOTION: TRANSCEIVER

RT-1161/FSS-9(V) AND PROCESSOR MX-9444/FSS-9(V)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers an ultrasonic motion signal transceiver and processor which combine to form an ultrasonic motion sensor, designed to detect the motion of an intruder inside a protected space.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

QQ-S-781	- Strapping, Steel, and Seals.
PPP-B-566	- Boxes, Folding, Paperboard.

FSC 6350

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-636	- Boxes, Shipping, Fiberboard.
PPP-B-676	- Boxes, Setup.

MILITARY

MIL-P-116	- Preservation-Packaging, Methods of.
MIL-C-45662	- Calibration System Requirements.
MIL-C-52853	- Control Unit, Alarm Set: C-9412/FSS-9(V).

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-454	- Standard General Requirements for Electronic Equipment.
MIL-STD-461	- Electromagnetic Interference Characteristics Requirements for Equipment.
MIL-STD-781	- Reliability Design Qualification and Production Acceptance Tests: Exponential Distribution.
MIL-STD-810	- Environmental Test Methods.
MIL-STD-1188	- Commercial Packaging of Supplies and Equipment

DRAWINGS

ME

TA13220E3201	- Processor, Ultrasonic Motion Signal - MX-9444/FSS-9(V).
TA13220E3211	- Circuit Card Assembly, Processor.
TA13220E3226	- Transceiver, Ultrasonic Motion Signal - RT-1161/FSS-9(V).

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TA13220E3233	- Circuit Card Assembly, Transceiver.
TA13220E3247	- Transducer Kit - Transmit, (White Dot).
TA13220E3248	- Transducer Kit - Receive, (Blue Dot).
C13220E2976	- Transducer.

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Description. The ultrasonic motion sensor (UMS) shall be one processor MS-9444/FSS-9(V), combined with a minimum of one to a maximum of twenty transceivers, RT-1161/FSS-9(V). The UMS shall be in accordance with Top Assemblies TA13220E3201 and TA13220E3226, and as specified herein.

3.1.1 Major components. The UMS shall consist of the following major components:

(a) Transceiver enclosure.
(b) Transceiver circuit card assembly TA13220E3233.
(c) Processor enclosure.
(d) Processor circuit card assembly TA13220E3211.
(e) Transducer Drawing C13220E2976.
(f) Transducer kit - transmit (white dot) TA13220E3247.
(g) Transducer kit - receive (blue dot) TA13220E3248.

When specified (see 6.2), any of the components or transducer kits listed above shall be furnished separately.

3.1.2 Function. The UMS shall detect movement within interior rooms by processing the Doppler frequency shifts between a transmitted and received ultrasonic signal. This frequency shift is caused by moving objects, with the volume influenced by the sensor. An alarm is generated when the criteria for detection are met. A UMS consisting of one processor and one transceiver, mounted in one corner of an unobstructed 30-foot by 20-foot rectangular room having a ceiling height of 8 to 12 feet, shall generate the alarm signal when movement meeting the criteria

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for alarm occurs in that room. Up to 20 transceivers can be used with the processor within a volume that is larger or contains obstructions. The UMS draws power from, and supplies alarm signals to, a control unit conforming to MIL-C-52853.

3.1.3 Drawings. The drawings forming a part of this specification are end product drawings. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Where tolerances could cumulatively result in incorrect fits, the contractor shall provide tolerances within those prescribed on the drawings to insure correct fit, assembly, and operation of the UMS. Any data (e.g. shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the contractor or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or his designated representative.

3.2 First article (preproduction model). The contractor shall furnish one or more UMS's and major components or transducer kits furnished separately for examination and testing within the time frame specified (see 6.2), to prove prior to starting production that his production methods will produce UMS's and major components or transducer kits furnished separately that comply with the requirements of this specification. Examination and tests shall be as specified in Section 4; and, unless otherwise specified herein, all examinations and tests shall be conducted by the contractor subject to surveillance and approval by the Government (see 6.3). When specified (see 6.2), the Government will conduct any or all of the preproduction examination and tests, as specified (see 6.2).

3.3 Initial production. When specified (see 6.2), the contractor shall furnish to the Government one or more UMS's for inspection as specified in 4.4.

3.4 Material. Material shall be as specified herein and on the applicable drawings. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification (see 6.9).

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3.5 Environmental conditions. The UMS and major components or transducer kits furnished separately shall withstand individual environmental conditions specified herein, or combinations thereof, without damage (see 6.4.3), performance degradation, or malfunction when tested in accordance with 4.6.2.5.

3.5.1 Operational.

(a) Temperature range: -20° F to +150° F.
(b) Humidity up to 95 percent relative.

3.5.2 Nonoperational and storage.

(a) Temperature range: -30° F to +165° F.
(b) Humidity up to 95 percent relative.
(c) Shock: 20 times the acceleration due to gravity (g's) for 11-milliseconds (ms) duration and bench handling.
(d) Vibration: Transportation conditions.

3.6 Detailed technical requirements.

3.6.1 Terminal board connections.

3.6.1.1 Transceiver terminal board. The transceiver terminal board shall be marked with identifying terminal numbers as follows:

Terminal Board 1 (TB1)	Function
1	Transmitter input signal
2	Transmitter input signal return
3	Receiver output signal
4	Receiver output signal return
5	Tamper signal
6	Tamper signal
7	Spare
8	Spare
9	Enclosure ground
10	Spare

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3.6.1.2 Signal processor terminal boards. The signal processor terminal boards shall be marked with identifying terminal numbers as follows:

Terminal Board 1 (TB1)	Function
1	Alarm output
2	Alarm output return
3	Spare
4	Spare
5	Tamper signal
6	Tamper signal
7	+20 volts direct current (Vdc) (from control unit)
8	20 Vdc return
9	Enclosure ground
10	Spare

On TB1, removable jumpers shall be provided across Terminals 2 and 3, 3 and 4, and 4 and 5.

Terminal Board 2 (TB2)	Function
1	Transmitter output signal
2	Transmitter output signal return
3	Receiver input signal
4	Receiver input signal return
5	Spare
6	Spare
7	Synchronization (sync) signal output
8	Signal ground
9	Enclosure ground
10	Spare

3.6.2 Functional characteristics.

3.6.2.1 Power. Power for an UMS shall be supplied to the processor from a 20 Vdc regulated power supply in the control unit. Reverse polarity connections to the processor shall not damage the UMS. An UMS

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consisting of any number of detectors from one to 20 shall operate as specified herein on regulated voltage between 18 and 22 Vdc with a current drain not exceeding 100 milliamperes (mA).

3.6.2.1.1 Transmitter power output. For an UMS consisting of a processor and a single transceiver, the on-axis sound pressure level at a distance of one foot from the transmit transducer shall not be less than 100 decibels (dB) above reference sound pressure level (RSPL). The frequency of the signal shall be 26.3 ±0.5 kilohertz (kHz).

3.6.2.2 Performance. The UMS shall be sensitive to broadband ultrasonic noise with frequency components within the passband of the UMS (25.3 to 27.3 kHz) that exceed the RSPL of 0.0002 microbars, at the receive transducer by 65 dB. The range of Doppler frequencies produced by unidirectional motion of targets with radial velocities between 0.4 feet per second (fps) and 20 fps for specified minimum durations. The specified minimum durations are 1 second at a velocity of 0.4 fps decreasing exponentially to 0.3 seconds at a velocity of 10 fps. For all velocities from 10 to 20 fps, the specified minimum durations are 0.3 seconds. The UMS shall perform as specified in 3.6.2.2.1 through 3.6.2.2.4 under the conditions listed below:

(a) Number of transceivers connected to the processor: up to 20.
(b) Cable between the processor and each transceiver: up to 500 feet of cable, RG 58/U or equivalent.

3.6.2.2.1 Alarm criteria for noise. The UMS shall be in the alarm state after broadband ultrasonic noise as specified below has been applied to one transceiver in a motion-free area for 5 seconds or more with its range control set to maximum and all other transceivers in an acoustically quiet, motion-free area. The noise shall have frequency components within the range 25.3 to 27.3 kHz that exceed the RSPL at the receive transducer by 65 dB.

3.6.2.2.1.1 Spectrum of broadband ultrasonic noise. The broadband ultrasonic noise generator shall be a band-limited white noise generator producing a uniform noise output in the 25-27 kHz band when applied to a suitable transducer (See Appendix Figure 6). The 3 dB bandwidth shall be 5-30 kHz with a 6 dB/octave slope outside this bandwidth.

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3.6.2.2.2 Alarm criteria for unidirectional motion. For a constant setting of the transceiver range control, the UMS shall be in the alarm state as listed in Table I upon receipt of the ultrasonic Doppler signals simulating the unidirectional motions listed in Table I with velocities in the range 0.55 to 17.6 fps. The UMS shall detect such motions in the presence of the ultrasonic clutter signal as listed in Table I. The UMS shall be in the non-alarm state when only the clutter signal is present and when neither the clutter signal nor the unidirectional Doppler signal is present. Unless otherwise specified herein, the performance described in this paragraph shall be obtained with only a single transceiver operating with a processor.

TABLE I. Schedule of UMS states for sums of simulated Doppler signals and clutter signals.

Simulated ultrasonic Doppler signal 1/			Ultrasonic clutter signal2/	UMS state
Type of motion	Target radial velocity4/	Target duration or period3/		
	fps			
Unidirectional	0.55	960 ms	None	Alarm
Unidirectional	1.1	780 ms	Present	Alarm
Unidirectional	2.2	570 ms	Present	Alarm
Unidirectional	4.4	380 ms	Present	Alarm
Unidirectional	8.8	310 ms	Present	Alarm
Unidirectional	17.6	300 ms	Present	Alarm
None	-	-	None	Non-alarm
None	-	-	None	Non-alarm
Oscillating	0.55	2 sec	None	Non-alarm
Oscillating	1.1	2 sec	Present	Non-alarm
Unidirectional	1.138	1 sec	None	Non-alarm
Unidirectional	35.2	1 sec	None	Non-alarm

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Notes: 1/The simulated ultrasonic Doppler signal shall be obtained from a Doppler signal synthesizer (see 6.4.4), and shall simulate the listed target motion. The simulated Doppler signal shall exceed the RSPL at the receive transducer by no more than 40 dB, unless otherwise noted.

2/The ultrasonic clutter signal shall consist of broadband ultrasonic noise. The clutter signal shall be continuous and shall exceed RSPL at the receive transducer by at least 45 dB.

3/The simulated ultrasonic Doppler signal for oscillatory motion shall be obtained from the Doppler signal synthesizer. The signal simulating oscillatory motion shall be:

(a) Continuous.
(b) Cyclic of period 2 seconds (sec).
(c) Zero at its start and symmetric about the zero reference.

4/For approaching and recording radial velocities of 17.6 fps, the simulated ultrasonic Doppler and clutter signals shall be applied to one transceiver, with 19 other transceivers located in an acoustically quiet area connected to the processor by up to 500 feet of RG 58/U cable.

The sound pressure level of the simulated ultrasonic Doppler signal at the receive transducer shall not exceed 55 dB above RSPL. The 19 transceivers and cable may be simulated by suitable electronic circuits having impedances and noise characteristics equivalent to those experienced in a typical string of 19 transceivers.

3.6.2.2.3 Oscillatory motion. For the minimum setting of the transceiver range control at which the UMS conforms to 3.6.2.2.2, the UMS shall be in the non-alarm state upon receipt of the ultrasonic Doppler signals simulating the oscillatory motions listed in Table I. The UMS shall be in the non-alarm state when simulated oscillatory motion appears in the presence of the ultrasonic clutter signal as listed in Table I.

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3.6.2.2.4 Out-of-band target signals. For the minimum setting of the transceiver range control at which the UMS conforms to 3.6.2.2.2, the UMS shall be in the non-alarm state upon receipt of the ultrasonic Doppler signals simulating the unidirectional motions listed in Table I with velocities less than 0.4 fps and more than 20 fps.

3.6.2.3 Detailed circuit requirements.

3.6.2.3.1 Transceiver range control. The transceiver range control shall provide an adjustment of at least 20 dB.

3.6.2.3.2 Alarm duration. When the alarm criteria are met, the UMS shall be in the alarm state for not less than 0.5 seconds nor more than 1.5 seconds after the alarm criteria are removed.

3.6.2.3.3 Alarm output. The resistance across the processor alarm output terminals (TB1-1 and TB1-2) shall be as follows:

(a) Non-alarm state: less than 150 ohms.
(b) Alarm state: more than 100 kilohms (k ohms).

The alarm output terminals shall be capable of accepting up to 10 mA dc in the non-alarm state and up to 30 Vdc in the alarm state.

3.6.2.3.4 Power interruption. When power on the UMS is interrupted, or the ground wire is disconnected, the UMS shall be in the alarm state.

3.6.2.4 Tamper protection. The resistance between terminals 5 and 6 of the detector and processor TB1 shall represent a tamper alarm condition by exceeding 100k ohms, or a non-tamper condition by being less than 150 ohms. The tamper alarm condition shall be presented whenever the cover is opened from 0.062 to 0.250 inch (flanges engaged) and shall continue until the cover is closed or the tamper switch actuator rod is pulled out. The switch shall reset by closing the cover.

3.6.2.5 Transceiver circuit card assembly. The transceiver circuit card assembly shall be in accordance with Top Assembly TA13220E3233 and have the following characteristics:

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(a) Receiver pre-Amp gain: For a sinusoidal signal of 26.3 ±0.1 kHz, 10.0 ±0.1 millivolt (mV) peak-to-peak (P-P) input signal, the output voltage at TB1-3 shall be 40 ±15 mV P-P with the range control set to Position 10 and shall not be greater than one-tenth of the voltage measured in the range control Position 10 with the range control set in Position 1.

(b) Continuity: Continuity (<10 ohms) shall be between the following terminals:

From	To
TB1-1	Transmit jack marked "RED".
TB1-2	Transmit jack marked "BLUE".
TB1-4	Receive jack marked "BLUE".
TB1-9	Chassis ground jack marked "BLACK".

(c) Tamper switch: The tamper switch shall meet the following continuity measurements as measured between TB1-5 and TB1-6; depress the tamper switch, <10 ohms; tamper switch released, >100k ohms; tamper switch in the "lock-out" detent, <10 ohms. When the transceiver circuit card assembly is mounted in the transceiver enclosure, the tamper switch shall meet the requirements of 3.6.2.4.

3.6.2.6 Processor circuit card assembly. The processor circuit card assembly shall be in accordance with Top Assembly TA13220E3211 and have the following characteristics:

(a) Alarm criteria for Doppler signals: For a Doppler-to-clutter ratio of -30 dB at point E3, per Figure 2, Appendix Test Method 7, and -67 ±3 dB at TB2-3, the alarm switch contacts, terminals 1 and 2 of TB1 shall indicate an alarm condition (>100k ohms).

(b) Alarm criteria for noise: For a single frequency input at TB2-3 with the plus side of C43 cliplead to the junction of R52 and R53, the alarm switch contacts, terminals 1 and 2 of TB1 shall indicate an alarm condition (>100k ohms).

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(c) Log amplifier gain criteria: The log amplifier shall produce an amplitude reduction of 0.1 ±0.02 V for each 10 dB step over a range of at least 40 dB. The initial starting voltage level is 0.8 volts at the "SIG" terminal on the processor circuit card assembly.

(d) Linear amplifier gain and Doppler filter criteria: The linear amplifier circuitry shall produce an amplitude reduction of a factor of 10 ±10% for each 20 dB step over a range of at least 40 dB. The initial starting voltage level is 1.4 ±0.1 V at the "NOISE" terminal of the processor. For a single frequency input of 800 Hertz (Hz) and 12 Hz input, a decrease by at least 3 dB shall be indicated at E3, per Figure 2 Appendix Test Method 7.

(e) Gating waveform criteria: The waveform present at the junction of R51 and R57 shall appear on the negative loop of the sinewave of the "NOISE" terminal for an input signal 200 Hz higher than the transmitter oscillator frequency and shift to the positive loop for an input signal 200 Hz lower than the transmitter oscillator frequency.

(f) Transmitter failure alarm criteria: The alarm switch contacts, terminals 1 and 2 of TB1 shall indicate an alarm condition (>100k ohms) when one of the frequency control transducer leads are lifted. The non-alarm condition shall be restored in at least 1.5 seconds.

(g) Tamper switch: The tamper switch shall meet the following continuity measurements as measured between TB1-5 and TB1-6; depress the tamper switch, <10 ohms; tamper switch release, >100k ohms; tamper switch in the "lock-out" detent, <10 ohms. When the processor circuit card assembly is mounted in the processor enclosure, the tamper switch shall meet the requirements of 3.6.2.4.

(h) Operating voltage criteria: The processor circuit card assembly shall operate from 20 ±2 Vdc at TB1-7 and TB1-8. TB1-7 shall be positive dc input and TB1-8 negative dc input. TP7 shall be 12 ±0.6 Vdc and TP8 shall be 6 ±0.6 Vdc with 20 Vdc applied. The input current shall be no greater than 40 mA. For reverse voltage input, the input current shall be no greater than 1 mA.

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3.6.2.7 Transducers. The transducers (red dot, blue dot, and white dot) shall be in accordance with Drawing C13220E2976 and have the following electrical characteristics:

(a) Red dot (C13220E2976-2):

Type: transmitter.
Minimum beamwidth at 3 dB points: 40°.
Excitation frequency: 26.30 ±0.15 kHz.
Excitation waveform: square wave, 20 volts P-P.
Minimum acoustic output: 26 dB relative to 1 microbar at 1 foot on axis.
Spurious emissions: shall emit no audible energy.

(b) Blue dot (C13220E2976-3):

Type: receiver.
Frequency of maximum acoustic-to-voltage conversion with 6.8k ohm resistor: 26.30 ±30 kHz.
Acoustic bandwidth at -3 dB points loaded with 6.8k ohm resistor: 1.2 kHz minimum.
Sensitivity: 58 dB relative to 1 volt per microbar.

(c) White dot (C13220E2976-1):

Type: transmitter.
Minimum beamwidth at 3 dB points: 40°.
Excitation frequency: 26.3 kHz.
Excitation waveform: Square wave, 20 volts P-P.
Minimum acoustic output: 26 dB relative to 1 microbar at 1 foot on axis.
Spurious emissions: shall emit no audible energy.

3.6.2.8 Transducer kit-transmit (white dot). The transducer kit-transmit (white dot) shall be in accordance with Top Assembly TA13220E3247. The transducer shall meet the requirements of 3.6.2.7 for white dot transducer.

3.6.2.9 Transducer kit-receive (blue dot). The transducer kit-receive (blue dot) shall be in accordance with Top Assembly TA13220E3248. The transducer shall meet the requirements of 3.6.2.7 for blue dot transducer.

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3.7 Electromagnetic susceptability. The electromagnetic susceptability characteristics of the UMS shall conform to MIL-STD-461, Methods RS03, CS01, CS02, and CS06, as modified in 3.7.1 and 3.7.2.

3.7.1 Radiated. Radiated susceptability shall be conducted in an electric field of 3 volts per meter (V/m) for both steady-state and keyed conditions of the EMI field generating equipment; this field applies to frequencies up to 500 megahertz (MHz). Above 500 MHz, the electric field shall be 1 V/m.

3.7.2 Conducted. Conducted interference signals shall be applied to both power and signal lines as follows:

(a) Continuous wave, 30 Hz to 400 kHz.
(b) Interference spike voltage, 10 spikes 5 to 10 microseconds wide at a repetition rate of 7 to 10 pulses per second on input leads only. Random phasing of the spikes with respect to the power line frequency shall be employed.

3.8 Reliability. The UMS shall demonstrate a Mean-Time-Between-Failure (MTBF) of 10,000 hours when tested as specified in 4.6.2.7.

3.9 Identification marking. The UMS shall be identified in accordance with MIL-STD-130.

3.10 Government-loaned property. Unless otherwise specified (see 6.2), the following property in the quantities indicated will be loaned by the Government (see 6.8) for testing major components or transducer kits furnished separately:

Item No.	Description	Identification	Quantity
1.	Processor, Ultrasonic Motion Signal; MX-9444/FSS-9(V)	TA13220E3201	1
2.	Transceiver, Ultrasonic Motion Signal; RT-1161/FSS-9(V)	TA13220E3226	1

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3.11 Interchangeability. All parts having the same part number shall be functionally and dimensionally interchangeable. Interchangeable parts are defined as two or more like parts possessing such functional and physical characteristics as to be equivalent in performance and durability, and capable of being exchanged one for the other without alteration of the parts themselves or of adjoining parts, except for adjustment, and without selection for fit or performance.

3.12 Workmanship. Workmanship shall be in accordance with MIL-STD-454, Requirement 9, except soldering. These requirements are applicable to wiring, welding, brazing, plating, riveting, finishes, machine operations, screw assemblies, and freedom of parts from burrs, sharp edges, or any other damage or defect that could make the part (or equipment) unsuitable for the purpose intended.

3.12.1 Soldering. All soldering shall be performed in accordance with MIL-STD-454, Requirement 5.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications, standards, and drawings, as applicable.

4.1.2 Disassembly inspection. Failure of any examination or test by the preproduction model shall be cause for disassembly, in the presence of a Government representative, of the preproduction model to the extent necessary to determine the cause of failure. Each disassembled part shall be examined in detail for compliance with this specification and referenced drawings in regard to materials, dimensions, tolerances, and

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workmanship. Parts not complying with such requirements shall be rejected and shall be cause for rejection of the preproduction model. Reassembly with replacement parts and retesting shall be the responsibility of the contractor. Government-loaned property shall be excluded from the requirements of this paragraph.

4.1.3 Parts and components. Parts and components detailed on the drawings shall be inspected in accordance with the supplementary quality assurance provisions (SQAP) shown on the drawings. The drawings specify the characteristics requiring SQAP inspection, the sampling plan, and the basis for acceptance and rejection (see 6.4.6 and 6.5).

4.2 Classification of inspections. The UMS shall be inspected as specified herein. Individual testing of major components installed in the UMS is not required. Major components or transducer kits furnished separately shall be individually inspected as specified herein. The inspection requirements specified herein are classified as follows:

(a) Preproduction inspection (see 4.3).
(b) Initial production inspection (see 4.4).
(c) Quality conformance inspection (see 4.5).
(d) Inspection comparison (see 4.7).
(e) Inspection of packaging (see 4.8).

4.3 Preproduction inspection.

4.3.1 Examination. The preproduction UMS and major components or transducer kits furnished separately shall be examined as specified in 4.6.1. Presence of one or more defects shall be cause for rejection or for performing the disassembly inspection specified in 4.1.2.

4.3.2 Tests.

4.3.2.1 Complete UMS. The complete UMS shall be subjected to the tests marked "X" in Column 1 of Table II. Individual testing of major components of the complete UMS is not required. Failure of any test shall be cause for rejection or for performing the disassembly inspection specified in 4.1.2.

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4.3.2.2 Major components or transducer kits furnished separately. Major components or transducer kits furnished separately shall be subjected to the tests marked "X" in Column 1 of Tables III through VII. Failure of any test shall be cause for rejection or for performing the disassembly inspection specified in 4.1.2.

4.4 Initial production inspection. When specified (see 3.3), one or more initial production UMS's will be selected at random by the Government from UMS's being produced by production tooling and will be examined as specified in 4.6.1 and subjected to the tests marked "X" in Column 2 of Table II to determine conformance to the requirements of this specification. The inspection will be performed by the Government at a site selected by the Government. Acceptance of an initial production UMS shall not exclude the remaining UMS's from the quality conformance inspection and acceptance provisions specified in Section 4. In addition to any test specified as part of the initial production test, the Government reserves the right to conduct any or all other tests contained in this specification as part of the initial production test and failure of such additional tests shall have the same effect as failure of those tests specified as initial production tests.

4.4.1 Inspection failure. Failure of an initial production UMS to meet any requirement specified herein during and as a result of the inspection and tests specified in 4.4 shall be cause for rejection of the initial production UMS and shall be cause for refusal by the Government to continue acceptance of production UMS's until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on UMS's previously accepted and produced under the contract. Any deficiencies found as a result of the initial production inspection will be considered prima facie evidence that all UMS's accepted prior to the completion of the initial production inspection are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

4.5 Quality conformance inspection.

4.5.1 Sampling. Sampling for tests of the complete UMS and major components or transducer kits furnished separately shall be in accordance with MIL-STD-105.

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4.5.2 Examination. Each UMS and each major component or transducer kit furnished separately shall be examined for major defects specified in 4.6.1. Presence of one or more defects shall be cause for rejection.

4.5.3 Tests.

4.5.3.1 Individual.

4.5.3.1.1 Complete UMS. Each complete UMS shall be subjected to the tests marked "X" in Column 3 of Table II. Individual testing of major components of the complete UMS is not required. Failure of any test shall be cause for rejection.

4.5.3.1.2 Major components or transducer kits furnished separately. Each major component or transducer kit furnished separately shall be subjected to the test marked "X" in Column 2 of Tables III through VII. Failure of the test shall be cause for rejection.

4.5.3.2 Samples.

4.5.3.2.1 Complete UMS. Complete UMS's shall be subjected to the tests marked "X" in Column 4 of Table II. Individual testing of major components of the complete UMS is not required. AQL shall be 2.5 percent defective.

4.5.3.2.2 Major components or transducer kits furnished separately. Major components or transuducer kits furnished separately shall be subjected to the tests marked "X" in Column 3 of Tables III through VII. AQL shall be 2.5 percent defective.

4.6 Inspection procedure.

4.6.1 Examination. The UMS and major components or transducer kits furnished separately shall be examined as specified herein for the following major defects:

101. Any part (or component) not in accordance with the SQAP requirements as shown on the drawings.
102. Components missing or not as specified.
103. Assembly not correct.
104. Electromagnetic interference suppression components and network not identical to the approved model.

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105. Identification or special marking missing, incomplete, or illegible.
106. Interchangeability not as specified.
107. Workmanship including wiring and soldering not as specified.

4.6.2 Tests.

4.6.2.1 Test conditions. Unless otherwise specified herein, tests shall be conducted at prevailing ambient temperatures and humidity conditions. Prior to testing, each UMS shall be subjected to a 72-hour burn-in during which the UMS shall be exercised through each of its functions not less than nine times during each 24 hours of burn-in. All failures shall be recorded and analyzed for cause of failure. Corrective action shall be taken when appropriate. Copies of these failure reports shall be forwarded to the contracting officer. UMS's for testing shall be selected from those UMS's having successfully completed the 72-hour burn-in.

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4.6.2.2 Test schedule.

TABLE II. Test schedule for complete UMS's.

Preproduction	Initial production	Quality conformance: Individual	Quality conformance: Sample	Tests	Requirement paragraph	Test paragraph
1	2	3	4	5	6	7
				Functional:		
X	X	X	-	(a) Power	3.6.2.1	4.6.2.4.1
X	X	X	-	(b) Performance	3.6.2.2	4.6.2.4.2
X	X	-	X	(c) Detailed circuit	3.6.2.3	4.6.2.4.3
X	X	-	X	(d) Tamper switch tolerance	3.6.2.4	4.6.2.4.4
X	X	-	X	Low temperature	3.5.1(a), 3.5.2(a)	4.6.2.5.1
X	X	-	X	High temperature	3.5.1(a), 3.5.2(a)	4.6.2.5.2
X	X	-	X	Humidity	3.5.1(b), 3.5.2(b)	4.6.2.5.3
X	X	-	-	Shock	3.5.2(c)	4.6.2.5.4
X	X	-	-	Vibration	3.5.2(d)	4.6.2.5.5
X	X	-	-	Electromagnetic susceptability	3.7	4.6.2.6
X	X	-	-	Reliability 1/	3.8	4.6.2.7

1/ When specified (see 6.2) preproduction reliability tests can be deleted at time of contract award based upon equipment performance, past history, and contractor past performance.

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TABLE III. Test schedule for transceiver circuit card assembly.1/

Preproduction	Quality conformance		Tests	Requirement paragraph	Test paragraph
	Individual	Sample			
1	2	3	4	5	6
X	X	-	Functional	3.6.2.5	4.6.2.4.5
X	-	X	Low temperature	3.5.1(a), 3.5.2(a)	4.6.2.5.1
X	-	X	High temperature	3.5.1(a) 3.5.2(a)	4.6.2.5.2
X	-	X	Humidity	3.5.1(b) 3.5.2(b)	4.6.2.5.3
X	-	-	Shock	3.5.2(c)	4.6.2.5.4
X	-	-	Vibration	3.5.2(d)	4.6.2.5.5

1/ Place transceiver circuit card assembly in Government-loaned transceiver enclosure or equivalent to simulate normal environment when installed in UMS or transceiver enclosure.

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TABLE IV. Test schedule for processor circuit card assembly.1/

Preproduction	Quality conformance		Tests	Requirement paragraph	Test paragraph
	Individual	Sample			
1	2	3	4	5	6
X	X	-	Functional	3.6.2.6	4.6.2.4.6
X	-	X	Low temperature	3.5.1(a), 3.5.2(a)	4.6.2.5.1
X	-	X	High temperature	3.5.1(a) 3.5.2(a)	4.6.2.5.2
X	-	X	Humidity	3.5.1(b), 3.5.2(b)	4.6.2.5.3
X	-	-	Shock	3.5.2(c)	4.6.2.5.4
X	-	-	Vibration	3.5.2(d)	4.6.2.5.5

1/ Place processor circuit card assembly in Government-loaned processor enclosure or equivalent to simulate normal environment when installed in UMS or processor enclosure.

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TABLE V. Test schedule for transducer. 1/

Preproduction	Quality conformance		Tests	Requirement paragraph	Test paragraph
	Individual	Sample			
1	2	3	4	5	6
X	X	-	Functional	3.6.2.7	4.6.2.4.7
X	-	X	Low temperature	3.5.1(a), 3.5.2(a)	4.6.2.5.1
X	-	X	High temperature	3.5.1(a), 3.5.2(a)	4.6.2.5.2
X	-	X	Humidity	3.5.1(b), 3.5.2(b)	4.6.2.5.3
X	-	-	Shock	3.5.2(c)	4.6.2.5.4
X	-	-	Vibration	3.5.2(d)	4.6.2.5.5

1/ Place transducer in Government-loaned UMS, transceiver enclosure, or equivalent to simulate normal environment when installed in UMS or transceiver enclosure.

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TABLE VI. Test schedule for transducer kit-transmit (white dot). 1/

Preproduction	Quality conformance		Tests	Requirement paragraph	Test paragraph
	Individual	Sample			
1	2	3	4	5	6
X	X	-	Functional	3.6.2.8	4.6.2.4.8
X	-	X	Low temperature	3.5.1(a), 3.5.2(a)	4.6.2.5.1
X	-	X	High temperature	3.5.1(a), 3.5.2(a)	4.6.2.5.2
X	-	X	Humidity	3.5.1(b), 3.5.2(b)	4.6.2.5.3
X	-	-	Shock	3.5.2(c)	4.6.2.5.4
X	-	-	Vibration	3.5.2(d)	4.6.2.5.5

1/ Place transducer kit-transmit (white dot) in Government-loaned UMS, transceiver enclosure, or equivalent to simulate normal environment when installed in UMS, or transceiver enclosure.

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TABLE VII. Test schedule for transducer kit-receive (blue dot).1/

Preproduction	Quality conformance		Tests	Requirement paragraph	Test paragraph
	Individual	Sample			
1	2	3	4	5	6
X	X	-	Functional	3.6.2.9	4.6.2.4.9
X	-	X	Low temperature	3.5.1(a), 3.5.2(a)	4.6.2.5.1
X	-	X	High temperature	3.5.1(a), 3.5.2(a)	4.6.2.5.2
X	-	X	Humidity	3.5.1(b), 3.5.2(b)	4.6.2.5.3
X	-	-	Shock	3.5.2(c)	4.6.2.5.4
X	-	-	Vibration	3.5.2(d)	4.6.2.5.5

1/ Place transducer kit-receive (blue dot) in Government-loaned UMS, transceiver enclosure, or equivalent to simulate normal environment when installed in UMS or transceiver enclosure.

4.6.2.3 Test methods. Tests shall be conducted in accordance with the test methods in the appendix and as specified herein. Calibration of test equipment shall be in accordance with MIL-C-45662.

4.6.2.4 Functional.

4.6.2.4.1 Power. The UMS shall be tested in accordance with Appendix Test Method 1. Nonconformance to 3.6.2.1 shall constitute failure of this test.

4.6.2.4.2 Performance. The UMS shall be tested in accordance with Appendix Test Method 2. Nonconformance to 3.6.2.2 shall constitute failure of this test.

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4.6.2.4.3 Detailed circuit. The UMS shall be tested in accordance with Appendix Test Method 3. Nonconformance to 3.6.2.3 shall constitute failure of this test.

4.6.2.4.4 Tamper switch tolerance. The UMS shall be tested in accordance with Appendix Test Method 4. Nonconformance to 3.6.2.4 shall constitute failure of this test.

4.6.2.4.5 Transceiver circuit card assembly. The transceiver circuit card assembly shall be tested in accordance with Appendix Test Method 6. Nonconformance to 3.6.2.5 shall constitute failure of this test.

4.6.2.4.6 Processor circuit card assembly. The processor circuit card assembly shall be tested in accordance with Appendix Test Method 7. Nonconformance to 3.6.2.6 shall constitute failure of this test.

4.6.2.4.7 Transducer. The transducer shall be tested in accordance with Appendix Test Method 8. Nonconformance to 3.6.2.7 shall constitute failure of this test.

4.6.2.4.8 Transducer kit-transit (white dot). The transducer kit-transmit (white dot) shall be tested in accordance with Appendix Test Method 8. Nonconformance to 3.6.2.8 shall constitute failure of this test.

4.6.2.4.9 Transducer kit-receive (blue dot). The transducer kit-receive (blue dot) shall be tested in accordance with Appendix Test Method 8. Nonconformance to 3.6.2.9 shall constitute failure of this test.

4.6.2.5 Environmental. The UMS and major components or transducer kits furnished separately shall be subjected to the following series of tests in accordance with MIL-STD-810. Prior to and following each test, conduct the operational tests in accordance with Appendix Test Method 5.

4.6.2.5.1 Low temperature. Conduct Test Method 502.1, Procedure I, storage temperature -30° F, for not less than 48 hours, equipment operating temperature -20° F. Major components or transducer kits furnished separately shall be subjected to the -20° F operational test. At the operating temperature perform the operating test in accordance with Appendix Test Method 5. The frequency measurement shall be within ±131 Hz of the recorded reading of Step 2(e), Appendix Test Method 5, measurements at room temperature (+75° F ±5° F) prior to this environmental test.

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4.6.2.5.2 High temperature. Conduct Test Method 501.1, Procedure II, except that the nonoperating temperature shall be +165° F; the operating temperature shall be +150° F. Major components or transducer kits furnished separately shall be subjected to the +150° F operational tests. At the operating temperature perform the operating test in accordance with Appendix Test Method 5. The frequency measurement shall be within ±131 Hz of the recorded reading of Step 2(e), Appendix Test Method 5, measurement at room temperature (+75° F ±5° F) prior to this environmental test.

4.6.2.5.3 Humidity. Conduct Test Method 507.1, Procedure I, with the following exceptions:

(a) Step 5 shall be for 5 cycles (not less than 120 hours). Conduct the operating tests in accordance with Appendix Test Method 5, at the completion of each cycle.
(b) During Steps 2 and 3, the relative humidity shall be maintained at 95 percent or greater.
(c) During Step 4, the relative humidity shall be maintained at 92 percent or greater.
(d) Measurement shall commence 1 hour and 15 minutes prior to the completion of each 24 hour cycle. For the tamper test, the covers of the UMS may be wiped of excess external moisture prior to the performance of this test. This test may be the last functional test performed prior to continuation of humidity cycling. The UMS and major components or transducer kits when furnished separately may be removed from the environmental chamber for measurement; however, unsuccessful completion of measurements or unsuccessful return of the UMS to the test chamber within 1 hour and 15 minutes subsequent to the initiation of measurements shall constitute failure of this test.

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4.6.2.5.4 Shock.

(a) Basic design: Conduct Test Method 516.2, Procedure I, equipment not operating, sawtooth shock pulse, peak value 20 g's, duration 11 ms.
(b) Bench handling: Conduct Test Method 516.2, Procedure V.

4.6.2.5.5 Vibration. Conduct Test Method 514.2, Procedure X, Curve AX, equipment not operating.

4.6.2.5.6 Failure criteria. Nonconformance to 3.5, or nonconformance to the requirement of Appendix Test Method 5 shall constitute failure of these tests.

4.6.2.6 Electromagnetic susceptability. The UMS shall be subjected to the tests of MIL-STD-461 as follows.

4.6.2.6.1 Radiation susceptability. Conduct Test Method RS03. Electric field requirements of RS03 shall be modified and limited as specified in 3.7.1. All leads (sensor, alarm and power) external to the signal processor enclosure, shall be exposed to the electric field. The leads shall not be less than 4 feet in length.

4.6.2.6.2 Conducted susceptability. Conduct test CS01 (except level shall be reduced 0.4 volts between 24.5 and 27.5 kHz), CS02, and CS06 as modified in 3.7.2.

4.6.2.6.3 Failure criteria. The contractor shall furnish the contracting officer a report of the tests required by MIL-STD-461. Disapproval of the report shall constitute failure of these tests.

4.6.2.7 Reliability. Using an upper test MTBF (Θ_0) of 10,000 hours and a lower test MTBF (Θ_1) of 5000 hours the UMS shall be tested as specified herein, with "accept" and "reject" criteria in accordance with MIL-STD-781, Test Plan VIIIC. Continue testing until either an "accept" or "reject" decision is reached. A reliability failure is defined as any malfunction which causes or may cause one or more of the following:

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inconsistencies exist between the written specification and

(a) Failure to commence operation, cessation of operation, or degradation of performance below the designated levels.
(b) Serious damage to the system.
(c) Serious personnel safety hazards.

4.6.2.7.1 Test procedures. The tests shall be conducted under ambient laboratory conditions, without temperature, humidity, or vibration cycling. Equipment shall be energized at the power input specified in 3.6.2.1. Operation shall be continuous throughout the test. The 5-second, wideband ultrasonic noise signal specified in 2.1(g) of Appendix Test Method 2, shall be applied to the UMS not less than nine times during each 24 hour period. At the end of the reliability test, the UMS shall be subjected to the tests specified in 4.6.2.4.1 and 4.6.2.4.2. The tests shall commence after installation and burn-in period of 72 hours specified in 4.6.2.1. Parts or components of known limited life, such as batteries, shall have a life stipulated prior to test. Failures of these parts occurring after the end of the stipulated period are nonrelevant, but any dependent failures caused thereby are relevant. Failures chargeable to workmanship deficiencies shall be counted as relevent. Alarms not deliberately excited as part of the equipment test routine shall be reported and cause of the alarm identified if practicable. This accounting shall be maintained separately from the failure accounting. Redundant elements, such as indicator lamps shall not be repaired nor counted as failure until the UMS has failed to perform its intended function. Preventive maintenance procedures specified for the equipment during normal operation shall be performed during the reliability test. No additional preventive maintenance is allowed during the test.

4.6.2.7.2 Failure criteria. Nonconformance to 3.8 shall constitute failure of this test.

4.7 Inspection comparison. The Government may select UMS's at any time during the contract production period and subject these UMS's to the examination and tests specified in 4.4 to determine conformance to the requirements of this specification. The inspection will be performed by the Government, at a site selected by the Government, on units selected at random from those which have been accepted by the Government, and will not include the previously inspected preproduction model and initial production UMS's. In addition to any test specified as part of

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the inspection comparison, the Government reserves the right to conduct any and all other tests contained in this specification as part of the inspection comparison and failure of such additional tests shall have the same effect as failure of those tests specified as inspection comparison.

4.7.1 Inspection failure. Failure of an inspection comparison UMS to meet any requirement specified herein during and as a result of the examination and tests specified in 4.7 shall be cause for rejection of the inspection comparison UMS and shall be cause for refusal by the Government to continue acceptance of production UMS's until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on UMS's previously accepted and produced under the contract. Any deficiencies found as a result of the inspection comparison will be considered prima facie evidence that all UMS's accepted prior to the completion of inspection comparison are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

4.8 Inspection of packaging.

4.8.1 Quality conformance inspection of pack.

4.8.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.8.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.8.1.3 Examination. Samples selected in accordance with 4.8.1.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

108.	Materials, methods, and containers not as specified for Level A or Level B. Each incorrect material, method, or container shall be considered one defect.
109.	Boxed contents not cushioned to prevent movement or damage as specified for Level A or Level B.

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110. Strapping not as specified for Level A.
111. Quantities packed together exceed the size or weight limitation of the box for Level A or Level B.
112. Marking missing, illegible, incorrect, or incomplete for Level A, Level B, or Commercial.

5. PACKAGING

5.1 Preservation. Preservation shall be Level A, Level B, or Commercial as specified (see 6.2).

5.1.1 Level A. The UMS components or transducer kits shall be preserved by the applicable following procedure, depending upon procurement of complete UMS's or procurement of major components or transducer kits furnished separately.

5.1.1.1 Complete UMS's. Each signal processor and each ultrasonic transceiver shall be preserved in accordance with MIL-P-116, Method IIc. The two components shall then be placed together in a close-fitting box conforming to PPP-B-636, W6c or W5c, style optional. The components shall be cushioned within the box to prevent movement or damage.

5.1.1.2 Major components or transducer kits furnished separately.

5.1.1.2.1 Signal processors and ultrasonic transceivers. Signal processors and ultrasonic transceivers shall be preserved as specified in 5.1.1.1 and individually placed in a box as specified in 5.1.1.1.

5.1.1.2.2 Enclosures. Each signal processor enclosure or each ultrasonic transceiver enclosure shall be individually preserved in accordance with MIL-P-116, Method IC-1 or IC-3.

5.1.1.2.3 Printed wiring board assemblies. Each printed wiring board assembly shall be preserved in accordance with MIL-P-116, Method IIe, and placed in an individual close-fitting box conforming to PPP-B-566, Variety 1, style, type, and class optional or PPP-B-676, Variety 1, style, type, and class optional.

5.1.1.2.4 Transducer kit. Each transducer kit shall be preserved as specified in 5.1.1.2.3.

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5.1.1.2.5 Consolidation. Major components of like description preserved as specified in 5.1.1.2.2, 5.1.1.2.3, and 5.1.1.2.4 shall be consolidated together in a close-fitting box conforming to PPP-B-636, W6c, or W5c, style optional in quantities not to exceed the weight limitation of the box. The contents shall be cushioned to prevent movement or damage.

5.1.2 Level B. Preservation shall be as specified in 5.1.1 for Level A, except Method IA-8 shall be used in place of Method IIc and Method IA-15 shall be used in place of Method IIe.

5.1.3 Commercial. Each complete UMS, major component, or transducer kit furnished separately, shall be preserved in accordance with MIL-STD-1188.

5.2 Packing. Packing shall be Level A, Level B, or Commercial as specified (see 6.2).

5.2.1 Level A. The UMS components shall be packed by the following applicable procedure, depending upon procurement of complete UMS's or procurement of major components or transductor kits separately.

5.2.1.1 Complete UMS's. Complete UMS's, preserved as specified in 5.1, shall be packed together in a close-fitting box conforming to PPP-B-601, Overseas Type, style optional, in quantities not to exceed the weight limitation of the box. Closure and strapping shall be in accordance with the appendix to the box specification. Strapping shall conform to QQ-S-781, Class 1, Type I or IV, size as applicable. Unless otherwise specified (see 6.2), strapping shall be Finish B. When specified (see 6.2), strapping shall be Finish A.

5.2.1.2 Major components or transducer kits. Major components or transducer kits furnished separately, preserved as specified in 5.1, shall be packed together as specified in 5.2.1. Only components of like description shall be packed together.

5.2.2 Level B. Complete UMS's or major components or transducer kits furnished separately, preserved as specified in 5.1, shall be packed as specified in 5.2.1.1 or 5.2.1.2 as applicable except boxes shall be Domestic Type. As an alternate, boxes conforming to PPP-B-636, V3c, V11c, V13c, or V15c may be used when the quantity to be packed does not exceed the gross weight or size limitation of the box. Box closure and strapping shall be in accordance with the appendix to the applicable box specification.

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5.2.3 Commercial. Complete UMS's, major components or transducer kits furnished separately preserved as specified in 5.1, shall be packed in accordance with MIL-STD-1188.

5.3 Marking.

5.3.1 Military. Marking for Level A or Level B shall be in accordance with MIL-STD-129 and the special marking specified in 5.3.3.

5.3.2 Commerical. Marking for Commercial packaging shall be in accordance with MIL-STD-1188 and the special marking specified in 5.3.3.

5.3.3 Special marking. Each shipping container shall be marked with code labels in accordance with MIL-STD-129. The labels shall contain the Project Code "JID".

6. NOTES

6.1 Intended use. The ultrasonic motion sensor is intended for use as a component of the Joint-Services Interior Intrusion Detection System (J-SIIDS).

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) When major components or transducer kits are to be furnished separately and number to be furnished (see 3.1.1).
(c) Time frame required for submission of preproduction model, and number of UMS's, major components, or transducer kits furnished separately required (see 3.2).
(d) When the Government will conduct any or all of the preproduction model examination and tests. When the Government will conduct some but not all of the preproduction examination and tests, the contracting officer should specify which examinations and tests will be conducted by the Government and which examination and tests shall be conducted by the contractor (see 3.2).

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(e) When initial production inspection is required and number of UMS's to be furnished, when applicable (see 3.3).
(f) When Government-loaned property is not required (see 3.10).
(g) When preproduction reliability tests may be deleted (see Table II).
(h) Degree of preservation and degree of packing required (see 5.1 and 5.2).
(i) When other than Finish B strapping is required (see 5.2.1).

6.3 Preproduction model. Any changes or deviations of production UMS's and major components or transducer kits furnished separately from the approved preproduction model during production will be subject to the approval of the contracting officer. Approval of the preproduction model will not relieve the contractor of his obligation to furnish UMS's and major components or transducer kits furnished separately conforming to this specification.

6.4 Definitions. The following definitions shall apply throughout this specification.

6.4.1 False alarm. An alarm generated by stimuli other than those for which the UMS is designed to alarm.

6.4.2 Tamper alarm. Alarm resulting from removal of the sensor assembly cover or pry-off of the sensor assembly from its mounting.

6.4.3 Damage. Damage is defined as deformation, corrosion, loosening of parts, breakage, change of fit or any component or part, or any physical condition resulting in nonconformance to the provisions of this specification.

6.4.4 Doppler signal synthesizer. For a given basic transmitted ultrasonic signal, this equipment produces an ultrasonic signal equal in frequency to the basic frequency plus or minus Doppler frequencies corresponding to selectable radial target velocities, and adjustable in amplitude to simulate target range. The output signal may, as desired, simulate a unidirectional target, or one oscillating in place; and may be either steady-state or pulsed, with the pulse duration selectable. The output signal may be hard-wired to the load or radiated (air-coupled) to the load by close transducer spacing.

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6.4.5 Wideband ultrasonic signal generator. The wideband ultrasonic signal generator (WUSG) shall produce wideband ultrasonic energy, continuous wave (CW) or pulsed and adjustable in amplitude. The output signal may be hard-wired to the load or radiated (air-coupled) to the load.

6.4.6 Supplementary quality assurance provisions (SQAP). A SQAP is a contractual requirement that supplements Section 4 of the specification. A SQAP indicates the minimum requirements which must be inspected on the product drawings to verify the design objectives of the product and assure interchangeability of the repair parts.

6.5 Supplementary quality assurance provisions (SQAP) records. The contracting officer should require the contractor to maintain records of all SQAP inspections. A suggested paragraph is as follows:

"The contractor shall maintain complete records of all examinations and tests performed to verify the requirements of classified SQAP characteristics. The records shall include, as a minimum, lot size, sample size, drawing requirement, actual measurement, number and type of deficiencies found, quantity approved, quantity rejected, and corrective action taken when applicable."

6.6 Data requirements. The contracting officer should include requirements for such data as technical publications, instructional materials, illustrated parts lists, and contractor's maintenance and operation manual to be furnished with each UMS.

6.7 Provisioning. The contracting officer should include provisioning requirements for repair parts and maintenance tools as necessary (including any special tools), and instructions on shipment of UMS's.

6.8 Government-loaned property. The contracting officer should arrange to loan the property specified in 3.10, when applicable (see 6.2).

6.9 Recycled material. It is encouraged that recycled material be used, when practical, as long as it meets the requirements of this specification (see 3.4).

Custodian:
Army - ME
Air Force - 99

Review activity:
DLA - GS

User activities:
Navy - SH, MC

Preparing activity:
Army - ME

Project 6350-A116

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APPENDIX

FOR

ULTRASONIC MOTION SIGNAL PROCESSOR

AND TRANSCEIVER

10. SCOPE

10.1 Scope. The test methods contained within this appendix are to determine whether ultrasonic motion signal sensors and major components procured under this specification conform to the requirements set forth.

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TEST METHOD 1

POWER TEST

1. TEST APPARATUS:

(a) Power supply, 20 ±2 Vdc, 100 mA.
(b) Ammeter, 100 mA.
(c) Digital voltmeter, dc.

2. TEST PROCEDURE:

(a) Place the processor in operation by applying 20 ±0.3 Vdc through the ammeter to terminals TB1-7 and TB1-8. The current shall not exceed 38 mA.
(b) Connect the 20 transceivers to TB2 of the processor. The total current shall not exceed 100 mA.
(c) Reverse the polarity on terminals TB1-7 and TB1-8 for 10 seconds. The current shall not exceed 1 mA.
(d) Reconnect as in Step (b). The current shall not exceed 100 mA.
(e) Measure the voltage between Test Point 7 (TP-7) (POSITIVE) and TB-3 (NEGATIVE) with a voltmeter. The voltage shall be 12 ±1 Vdc.
(f) Vary the voltage at TB1-7 and TB1-8 from 18 to 22 V. The voltage between TP-7 and TP-3 (12 ±1 Vdc) shall not vary more than ±100 mV.

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TEST METHOD 2

PERFORMANCE TEST

1. TEST APPARATUS:

(a) Power supply, 20 ±2 Vdc, 100 mA.
(b) Power supply, 30 ±2 Vdc, 20 mA.
(c) Signal generator producing ultrasonic broadband noise, of variable amplitude, hard-wire or radiated (See Figure 6).
(d) Sound level meter with microphone, measures radiated ultrasonic energy, calibrated in sound pressure level (SPL).
(e) Oscilloscope.
(f) Doppler signal synthesizer.

2. TEST PROCEDURE:

2.1 Alarm criteria for noise.

(a) Place 19 of 20 transceivers in an acoustically quiet, motion-free area. One test transceiver shall be excited by the synthesizer test signals.
(b) Connect all 20 transceivers to TB2 of the processor by 500 feet of cable. Place the sound level meter microphone in close proximity to the test transceiver receive transducer.
(c) Set all transceiver range controls to Position 10.
(d) Place the UMS in operation by applying 20 ±0.3 Vdc to processor terminals TB1-7 and TB1-8.
(e) Operate the WUSG to radiate broadband ultrasonic noise to the microphone and test transceiver. With the sound level meter passband adjusted for the frequency octave centered at 31.5 kHz, adjust the WUSG output level to read 65 dB SPL (above 0.0002 microbar) on the sound level meter.
(f) Turn off the ultrasonic noise and momentarily short-circuit capacitor C54 (56 microfarads) to reset the noise detector.
(g) Turn on the ultrasonic noise for 5 seconds. The UMS shall alarm.

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2.2 Alarm criteria for unidirectional motion.

2.2.1 Target signals and background clutter.

(a) With the processor and transceiver interconnected as in 2.1(b), air-couple the transceiver transmit transducer to the Doppler signal synthesizer input.
(b) Hard-wire the WUSG output to be summed with the Doppler signal synthesizer output, prior to the signal synthesizer air-coupled output.
(c) Apply 30 +0, -0.3 Vdc, through 2.7k ohms, to processor terminals TB1-1 and TB1-2, and connect oscilloscope across these terminals to monitor alarm function.
(d) With UMS in operation as in 2.1(d), air-couple (closely spaced) summed WUSG and synthesizer output signals to sound level meter microphone.
(e) With Doppler signal synthesizer turned off, operate WUSG to produce broadband ultrasonic noise. Adjust WUSG output level to obtain a reading of 45 dB SPL on the sound level meter. This broadband noise level shall be used in all subsequent steps involving severe clutter.
(f) With the WUSG turned off, operate the Doppler signal synthesizer in the steady-state mode, at 0.55 fps Adjust the signal synthesizer output level to obtain a reading of 40 dB SPL on the sound level meter (worst-case target return). This target return signal level shall be used in all subsequent steps involving target signals.
(g) Remove the air-coupled, summed signal-and-clutter output from the sound level meter microphone and apply it to the transceiver receive transducer.
(h) Turn off the WUSG, operating the Doppler signal synthesizer in the pulsed mode; apply a 0.55 fps target signal (no clutter) for a duration of 960 ms; the UMS shall alarm. Adjust the transceiver range control to the lowest setting at which an alarm can be obtained.
(i) Turn on the WUSG. Pulsing the synthesizer, apply 1.1 fps target signal (severe clutter) for a duration of 780 ms. The UMS shall alarm. If necessary, increase the range control to the setting required to obtain an alarm. Do not decrease the range control setting.

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(j) Repeat Step (i) for a 2.2 fps target signal and a duration of 570 ms.
(k) Repeat Step (i) for a 4.4 fps target signal and a duration of 380 ms.
(l) Repeat Step (i) for an 8.8 fps target signal and a duration of 310 ms.
(m) Increase Doppler signal synthesizer output level to 55 dB above RSPL. Repeat Step (i) for a 17.6 fps target signal and a duration of 300 ms. The alarm duration (as indicated on the oscilloscope), shall be from 0.5 to 1.5 seconds.
(n) Repeat Step (m) for -17.6 fps target signal.

2.2.2 Background clutter with no target signals.

(a) Having performed 2.2.1, leave the transceiver range control at the position to which it was set at the end of 2.2.1(n).
(b) With the same test equipment configuration as in 2.2.1(n) and the same signal levels as in 2.2.1(l), turn off the WUSG. The UMS shall not alarm.

2.2.3 Oscillating targets.

(a) Same as in 2.2.2(a).
(b) With the same test equipment configuration and signal levels as in 2.2.1, turn off the WUSG.
(c) Operate the Doppler signal synthesizer in the steady-state oscillating mode, at 0.5 Hz, for a target velocity of 0.55 fps (no clutter). The UMS shall not alarm.
(d) Turn on the WUSG. Operate the signal synthesizer in the steady-state oscillating mode, at 0.5 Hz, for a target velocity of 1.1 fps (severe clutter). The UMS shall not alarm.

2.2.4 Out-of-band target signals.

(a) Same as 2.2.2(a).
(b) Same as 2.2.3(b).
(c) Operate the signal synthesizer in the pulsed mode to apply a 0.138 fps target signal (no clutter) for a duration of 1 second. The UMS shall not alarm.

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(d) Repeat Step (c) with a 35.2 fps target signal. The UMS shall not alarm.

2.2.5 Transmitter power output.

(a) With one transceiver connected to a processor, measure transmitter output using the sound level meter. Place meter microphone 1 foot away from transmitter transducer. Sound level shall be equal or greater than 100 dB above RSPL.

Note: To minimize error in meter indication due to ambient noise, the UMS shall incorporate an external narrow band filter of ±5 kHz bandwidth centered at 26.3 kHz, or shall be measured using sound pressure meter with octave bandpass filter centered at 31.5 kHz.

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TEST METHOD 3

DETAILED CIRCUIT TEST

1. TEST APPARATUS:

(a) Power supply, 20 ±2 Vdc, 100 mA.
(b) Power supply, 30 ±2 Vdc, 20 mA.
(c) Voltmeter, ac, Hewlett-Packard Model 403B or equivalent.
(d) Doppler signal synthesizer.
(e) Oscilloscope.
(f) Ammeter, 30 mA.
(g) Voltmeter, dc.
(h) Multimeter, Simpson Model 260 or equivalent.

2. TEST PROCEDURES:

2.1 Transceiver range control.

(a) With processor and transceiver interconnected, air-couple the transceiver transmit transducer to the Doppler signal synthesizer input, and air-couple the signal synthesizer output to the transceiver receive transducer. Set the transceiver range control to maximum (Position 10).
(b) Monitor the processor "NOISE" test point with ac voltmeter.
(c) Place the UMS in operation by applying 20 ±0.3 Vdc to terminals TB1-7 and TB1-8.
(d) Apply a steady-state synthesized 4.4 fps target signal, at a level that produces a Doppler signal of 2.5 V rms at the "NOISE" test point.
(e) Turn the transceiver range control to minimum (Position 1). The signal level at the "NOISE" test point shall decrease by at least 20 dB.

2.2 Alarm duration.

(a) With equipment connected as in Step 2.1(a), apply 30 +0, -0.3 Vdc, through 2.7k ohms, to processor terminals TB-1 and TB-2 and connect oscilloscope across these terminals.

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(b) With transceiver range set to Position 7, and UMS in operation, apply a pulsed synthesized 4.4 fps target signal, for a duration of 500 ms, at the level in Step 2.1(d).
(c) The oscilloscope shall indicate an alarm. The alarm duration shall be from 0.5 to 1.5 seconds.

2.3 Alarm output.

(a) With equipment connected as in Step 2.1(a), apply 30 +0, -0.3 Vdc, through 2.7k ohms and the ammeter, to processor terminals TB1-1 and TB1-2 and connect dc voltmeter across these terminals.
(b) Repeat Step 2.2(b).
(c) In no-alarm state, ammeter shall indicate at least 10 mA, demonstrating 10 mA capability and resistance less than 150 ohms.
(d) Momentarily interrupt 20 Vdc power, causing UMS to alarm. The dc voltmeter shall read 30 +0, -1.2 Vdc, demonstrating: 30 Vdc hold and, resistance greater than 100k ohms.

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TEST METHOD 4

TAMPER SWITCH TOLERANCE TEST

1. TEST APPARATUS:

(a) Ohmmeter.
(b) Test covers, with hole for meter lead access, for processor and transceiver.

2. TEST PROCEDURE:

2.1 Processor.

(a) Connect an ohmmeter between the processor tamper terminals TB1-5 and TB1-6.
(b) Place the test cover on the processor. The resistance across tamper terminals TB1-5 and TB1-6 shall be less than 10 ohms.
(c) Raise the cover of the processor more than 0.062 inch but less than 0.250 inch (flanges engaged). The resistance between TB1-5 and TB1-6 shall increase to greater than 100k ohms.
(d) The resistance between TB1-5 and TB1-6 shall be greater than 100k ohms when the cover is removed.
(e) Pull the tamper switch fully out and lock. The resistance between TB1-5 and TB1-6 shall decrease to less than 10 ohms.

2.2 Transceiver.

(a) Connect an ohmmeter to transceiver tamper terminals TB1-5 and TB1-6.
(b) Place the test cover on the transceiver. The resistance across tamper terminals TB1-5 and TB1-6 shall be less than 10 ohms.
(c) Raise the cover on the transceiver more than 0.062 inch but less than 0.250-inch (flanges engaged). The resistance across the TB1-5 and TB1-6 shall increase to greater than 100k ohms.

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(d) The resistance between TB1-5 and TB1-6 shall be greater than 100k ohms when the cover is removed.
(e) Pull the tamper switch fully out and lock. The resistance between TB1-5 and TB1-6 shall decrease to less than 10 ohms.
(f) Repeat Steps (a) through (e) for each transceiver.

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TEST METHOD 5

OPERATIONAL TEST

1. TEST APPARATUS:

   (a) Power supply, 20 ±2 Vdc, 100 mA.
   (b) Power supply, 30 ±2 Vdc, 20 mA.
   (c) Oscilloscope.
   (d) Frequency counter, to 30 kHz.
   (e) Voltmeter, ac.
   (f) Doppler signal synthesizer.

2. TEST PROCEDURE:

   (a) With processor and transceiver interconnected, hard-wire Doppler signal synthesizer input to transceiver terminals TB1-1 and TB1-2 and monitor this connnection with frequency counter. Set transceiver range control to Position 7.
   (b) Hard-wire signal synthesizer output, through a 0.1 microfarad capacitor, to the preamplifier input (transceiver-red and blue pins). Disconnect receive transducer.
   (c) Apply 30 +0, -0.3 Vdc, through 2.7k ohms, to processor terminals TB1-1 and TB1-2, and connect oscilloscope across these terminals.
   (d) Place UMS in operation by applying 20 ±0.3 Vdc to processor terminals TB1-7 and TB1-8.
   (e) Measure transmit frequency with counter.
   (f) Monitor "NOISE" test point with ac voltmeter and operate Doppler signal synthesizer in steady-state mode at 4.4 fps target velocity. Adjust signal synthesizer output level to obtain Doppler signal of 2.5 V rms at the "NOISE" test point. This synthesizer output level shall be used in Steps (g) and (h).
   (g) Operate Doppler signal synthesizer in pulsed mode, and apply a 4.4 fps signal for a 400 ms duration. The UMS shall alarm. The alarm duration (as indicated on the oscilloscope) shall be from 0.5 to 1.5 seconds.
   (h) Operate signal synthesizer in steady-state oscillating mode, at 0.5 Hz, for a target velocity of 1.1 fps. The UMS shall not alarm.

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TEST METHOD 6

TRANSCEIVER CIRCUIT CARD ASSEMBLY

(TA13220E3233)

1. TEST APPARATUS:

(a) Power Supply, Hewlett-Packard Model 6202B or equivalent.
(b) Oscilloscope, Tektronix, Type 465 or equivalent.
(c) Signal Generator, Wavetek, Model 110 or equivalent.
(d) Ohmmeter, Simpson 260 or equivalent.
(e) Resistor 6.8k ±5%.
(f) Capacitor, 0.15 microfarad ±0.5%.
(g) Voltmeter dc, (M1), Data Precision Model 245 or equivalent.
(h) Test fixture (see Figure 1).

2. TEST PROCEDURE (ROOM TEMPERATURE):

2.1 Receiver Pre-Amp Gain.

(a) Arrange the equipment as shown in Figure 1. Adjust the dc power supply as necessary to obtain a voltage of 7.0 ±0.1 V, as shown on M-1.
(b) Apply a sinusoidal signal of 26.3 ±0.1 kHz, 10.0 ±0.1 mV peak-to-peak, through a 0.15 microfarad dc blocking capacitor to the pre-amp input terminal as shown. Note that a negative dc bias (-0.29 vdc, variable) will develop at the red input terminal during normal operation. Avoid connecting any "low pass" or low impedance devices to this terminal.
(c) With the range control set to Position 10, measure the pre-amp output voltage at TB1-3. This voltage shall be 40 ±15 mV peak-to-peak.
(d) With the range control set to Position 1, measure the pre-amp output voltage at TB1-3. The voltage shall be not greater than one-tenth of the voltage obtained in Step (c).

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(e) Turn the range control through Positions 1 through 10 several times. Insure that the output voltage response observed as in (c) and (d) is smooth and continuous and that any particular setting will result in a consistently repeatable output voltage.

2.2 Continuity Tests.

(a) With an ohmmeter, check for continuity (<10 ohm) between the following terminals. Note that the identification of the receive and transmit transducer terminals may not be marked on the printed circuit board. Refer to Figure 1, which shows the correct location.

From	To
TB1-1	Transmit jack marked "RED"
TB1-2	Transmit jack marked "BLUE"
TB1-4	Receive jack marked "BLUE" (board ground)
TB1-9	Chassis grounding jack marked "BLACK"

2.3 Tamper Switch.

(a) Connect an ohmmeter to the tamper terminals TB1-5 and TB1-6 of the transceiver card.
(b) Depress the tamper switch. The resistance across TB1-5 and TB1-6 shall be <10 ohms.
(c) Release the tamper switch. The resistance across the tamper terminals shall be >100k ohms.
(d) Pull the tamper switch fully out and lock. The resistance between TB1-5 and TB1-6 shall be <10 ohms.

3. TEST PROCEDURE (-20° F):

Procedures and specifications are identical to Paragraph 2 except circuit assembly in environmental chamber shall be "soaked" at -20° F for at least 1 hour before testing begins.

4. TEST PROCEDURE (+150° F):

Procedures and specifications are identical to Paragraph 2 except circuit assembly in environmental chamber shall be "soaked" at +150° F for at least 1 hour before testing begins.

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b. Any part out from ends which does not approximate

TEST METHOD 7

PROCESSOR CIRCUIT CARD ASSEMBLY

(TA13220E3211)

1. TEST APPARATUS:

(a) Power supply, 20 ±2 Vdc, 100 mA.
(b) Signal generator, Hewlett-Packard, Model 200CD or equivalent.
(c) Oscilloscope, Tektronix Type 465 or equivalent.
(d) Electronic counter, Hewlett-Packard Model 5300A/5304A or equivalent.
(e) Step attenuator, Hewlett-Packard Model 355B or equivalent (2 required).
(f) Voltmeter, ac, Hewlett-Packard Model 403B or equivalent.
(g) Multimeter, Simpson Model 260 or equivalent.
(h) Test fixture (See Figure 2).

2. TEST PROCEDURE (ROOM TEMPERATURE):

2.1 Input power.

(a) Arrange equipment as shown in Figure 2.
(b) Apply 20 ±0.3 Vdc through the ammeter to terminals TB1-7 and TB1-8. The current shall not exceed 40 mA.
(c) Reverse the polarity on terminals TB1-7 and TB1-8 for 10 seconds. The current shall not exceed 1 mA.
(d) Reconnect as in Step (b). The current shall not exceed 40 mA.
(e) Measure the voltage at pin 2 of Z5 with a dc voltmeter. The voltage shall be 12 ±0.6 Vdc.
(f) Measure the voltage to the plus side of C43 with a dc voltmeter. The voltage shall be 6 ±0.6 Vdc.

2.2 Alarm criteria for Doppler Signals.

(a) Connect the equipment per the arrangement shown in Figure 2. The transmitting transducer load for the transmitter output is simulated by the two 2k ohm and one 50 ohm resistors in series. The junction of the 50 ohm and one of the 2k ohm resistors is grounded to signal ground to preserve the balanced arrangement of

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the transmitter output. The 50 ohm resistor provides a 50 ohm source for driving the 50 ohm attenuator system. The second 2k ohm resistor serves as a summing resistor to provide a sample of the transmitted signal. The sampled transmitted signal is used for backscatter to the receiver. The 22k ohm resistor provides a means of applying the signal generator output to the summing junction to represent the Doppler signal.

(b) Measure the amplitude of the transmitter output at the input to the 2k ohm summing resistor (point E1 on Figure 2). The amplitude shall be 5 volts ±3 dB. Connect the voltmeter to the signal generator output (E2). Set the frequency in the range of 25 kHz and adjust the amplitude to a value corresponding to one-third of that measured for the transmitter output. These values of voltages and summing resistors yield a Doppler-to-clutter ratio of -30 dB at point E3. Connect the counter to the "NOISE" terminal of the processor printed circuit board. Adjust the frequency of the signal generator until a value of several hundred Hz is read on the counter.

(c) Connect the multimeter (ohms) to terminals 1 and 2 of (TB1), the alarm switch contacts. The unit shall indicate an alarm condition (contacts open) when the attenuators are set for a total of 37 ±3 dB. (Allow time for system to stabilize because of long time constants when large steps in attenuation are made.)

2.3 Alarm criteria for noise.

(a) Configure the equipment as described in 2.2(a) and (b). The "noise" for this test will be represented by a single frequency as provided by the signal generator. The noise function is evaluated under these conditions by deactivating the Doppler processing channel of the processor. The Doppler channel is deactivated by connecting a clip lead from the plus side of C43 (6 V supply) to the junction of R52 and R53.

(b) The unit shall indicate an alarm condition (contacts open) when the attenuators are set for a total of 25 ±3 dB. (Allow time for system to stabilize because of long time constants when large steps in attenuation are made.)

(c) An alarm condition shall be represented by an ohmmeter reading of greater than 100k ohms. A non-alarm condition shall be represented by an ohmmeter reading of no greater than 100 ohms.

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2.4 Log amplifier gain criteria.

(a) Configure the equipment as in 2.2(a) and (b).
(b) Connect the ac voltmeter to the "SIG" terminal on the processor printed circuit board. Adjust attenuator for an indication of 1.0 V rms.
(c) Remove 30 dB of attenuation. Meter shall indicate a value of 2.2 V rms ±20%.

2.5 Linear amplifier gain and Doppler filter criteria.

(a) Configure the equipment as in 2.2(a) and (b).
(b) Connect the ac voltmeter to the "NOISE" terminal on the processor printed circuit board. Set the attenuator at 30 dB. The voltmeter shall indicate 1.4 V ±4 dB.
(c) Increase the attenuator setting in 20 dB steps. The meter shall indicate an amplitude reduction of a factor of 10 ±10% for each 20 dB reduction over at least a 40 dB range.
(d) Return the attenuator settings to a total of 40 dB. Connect the counter to the "NOISE" terminal (along with the voltmeter). Confirm that the signal is in the range of several hundred Hz. Note the reading on the voltmeter. Adjust the oscillator to yield a frequency of 800 Hz on the counter. The meter indication shall have decreased by at least 3 dB at this frequency. Readjust the oscillator to yield a frequency of 12 Hz. The meter indication shall have decreased by at least 3 dB at this frequency.

2.6 Gating waveform criteria.

(a) Configure the equipment as in 2.2(a) and (b).
(b) Connect Channel 1 of the oscilloscope to the junction of R51 and R57. Connect Channel 2 to the "SIGNAL" terminal.
(c) Channel 1 shall indicate a pulse of 110 microseconds ±5% duration and 12 V ±10% amplitude. Switch on Channel 2 of the oscilloscope (dual trace, chopped mode operation). Adjust the signal generator to a frequency which is approximately 200 Hz higher than the transmitter oscillator.

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The pulse observed on Channel 1 shall occur at the peak of the negative loop of the sine wave observed on Channel 2. Adjust the signal generator to a frequency which is approximately 200 Hz lower than the transmitter oscillator. The pulse observed on Channel 1 shall occur at the peak of the positive loop of the sine wave observed on Channel 2.

2.7 Transmitter failure alarm test.

(a) Configure the equipment as in 2.2(a) and (b).
(b) Adjust the attenuator for at least 70 dB. The processor shall present a non-alarm condition.
(c) Momentarily lift one of the connectors for the frequency control transducer. The processor shall present an alarm condition.
(d) Restore the connection to the frequency control transducer. The non-alarm condition shall be restored in no more than 1.5 seconds.

2.8 Tamper switch.

(a) Connect an ohmmeter to tamper terminals TB1-5 and TB1-6 of the processor card.
(b) Depress the tamper switch. The resistance across tamper terminal TB1-5 and TB1-6 shall be $<$100 ohms.
(c) Release the tamper switch. The resistance across the tamper terminals shall be $>$100k ohms.
(d) Pull the tamper switch fully out and look. The resistance between terminals TB1-5 and TB1-6 shall be $<$100 ohms.

3. TEST PROCEDURE (-20° F):

Procedure and specifications are identical to Paragraph 2 except circuit assembly in environmental chamber shall be "soaked" at -20° F for at least 1 hour before test begins.

4. TEST PROCEDURE (+150° F):

Procedures and specifications are identical to Paragraph 2 except circuit assembly in environmental chamber shall be "soaked" at +150° F for at least 1 hour before test begins.

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TEST METHOD 8

TRANSDUCER

(C13220E2976-1, -2, and -3)

1. TEST APPARATUS:

(a) Power supply, Hewlett-Packard Model 6202B or equivalent.
(b) Anechoic chamber, 3 ft. x 3 ft. x 3 ft.
(c) Measuring amplifier, B&K type 2606 w/Type 4135 microphone and Type 2618 preamplifier or equivalent.
(d) Signal generator, Hewlett-Packard, Model 3310A or equivalent.
(e) Frequency counter, Hewlett-Packard, Model 5245L or equivalent.
(f) Spectrum analyzer, Hewlett-Packard, Model 3580A or equivalent.
(g) Attenuators, Hewlett-Packard, Models 355C and 355D or equivalent.
(h) Test fixtures (see Figures 3, 4, and 5).

2. TEST PROCEDURE:

2.1 Transducer (13220E2976-2) red dot.

(a) Arrange the equipment as shown in Figure 3.
(b) Calibrate the measuring amplifier in accordance with manufacturer's procedure.
(c) Set the measuring amplifier to provide an indication of $\pm$26 dB.
(d) Connect the output from the Test Schematic "A" to the transducer under test.
(e) Minimum acoustic output shall be +26 dB relative to 1 microbar at 1 foot on axis.

2.2 Transducer (13220E2976-1) white dot.

(a) Arrange the equipment as shown in Figure 4.
(b) Calibrate the measuring amplifier in accordance with manufacturer's procedure.

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(c) Set measuring amplifier to provide an indication of +20 dB.
(d) Connect the output from the signal generator to the transducer under test.
(e) Minimum acoustic output shall be +20 dB relative to 1 microbar at 1 foot on axis.

2.3 Transducer (13220E2976-3) blue dot.

(a) Arrange the equipment as shown in Figure 5.
(b) Calibrate the measuring amplifier in accordance with manufacturer's procedure.
(c) Check calibration of spectrum analyzer.
(d) Set spectrum analyzer controls as indicated below:

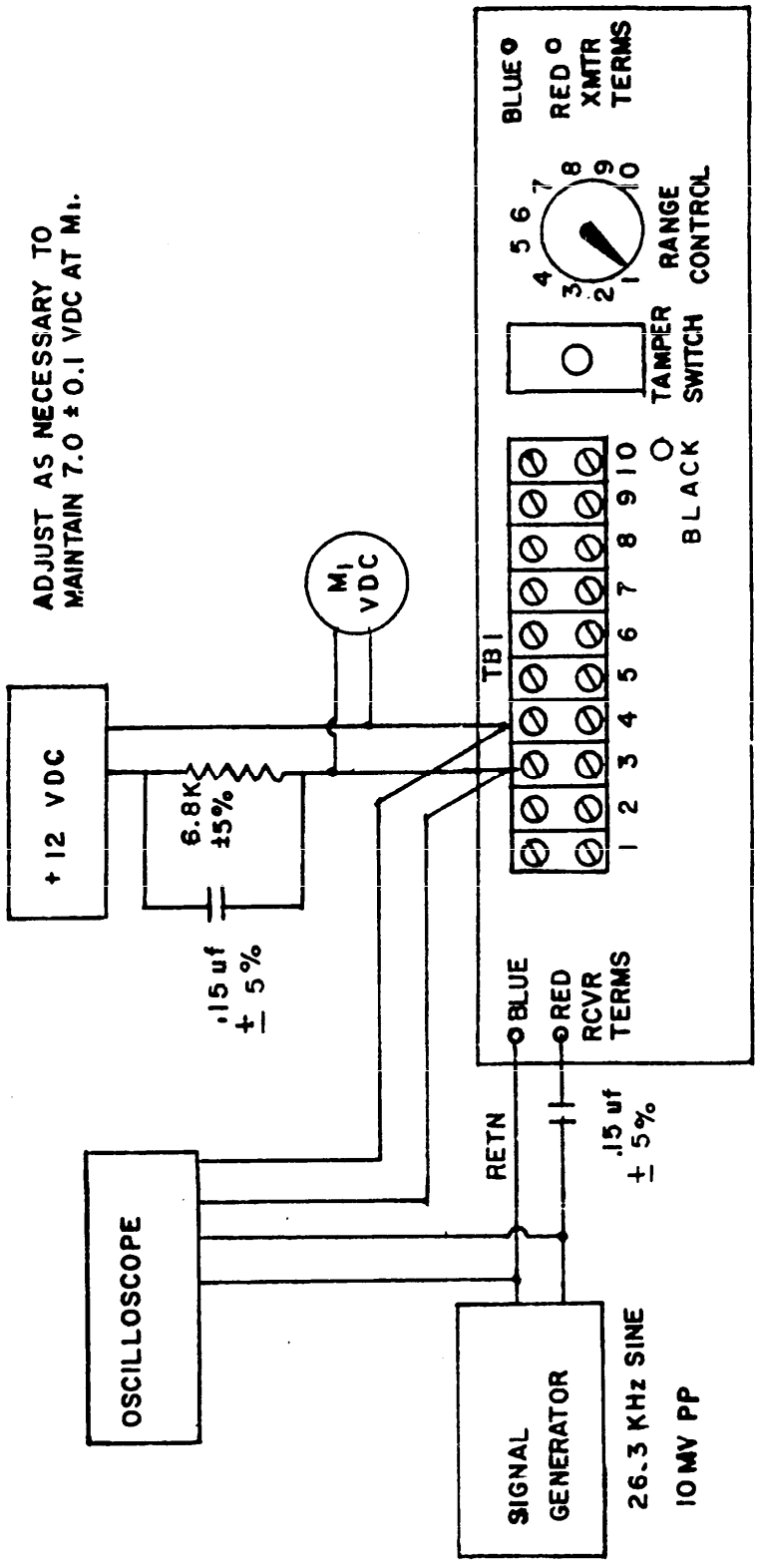
(1) Amplitude mode: log 10 dB/Div.
(2) Amplitude reference level: normal.
(3) Input sensitivity: +30.
(4) Resolution bandwidth: 100.
(5) Sweep time/Div.: 0.5 sec.
(6) Sweep mode: REP.
(7) Frequency Span/Div.: 0.
(8) Frequency: 26.3 kHz.

(e) Adjust the tracking oscillator level control of the spectrum analyzer to indicate 75 dB SPL on the measuring amplifier scale.
(f) Set the measuring amplifier input attenuator to 10 volts. Adjust the measuring amplifier gain control until the line on the spectrum analyzer is at -58 dB.
(g) Set the spectrum analyzer frequency control to 20.0 kHz and the frequency span/Div. to 1 kHz. Observe and store the sweep on the spectrum analyzer.
(h) Reset the input sensitivity of the spectrum analyzer to -10 dB. The system is now calibrated.
(i) Disconnect the microphone with preamplifier from the input of the measuring amplifer.
(j) Disconnect the input to the spectrum analyzer from the measuring amplifier and connect the output from the tranducer under test to the input of the spectrum analyzer. Connect a 6.8k ohm ±5% resistor across the transducer terminals.

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(k) Minimum acoustic bandwidth at the 3 dB points shall be 1.2 kHz.

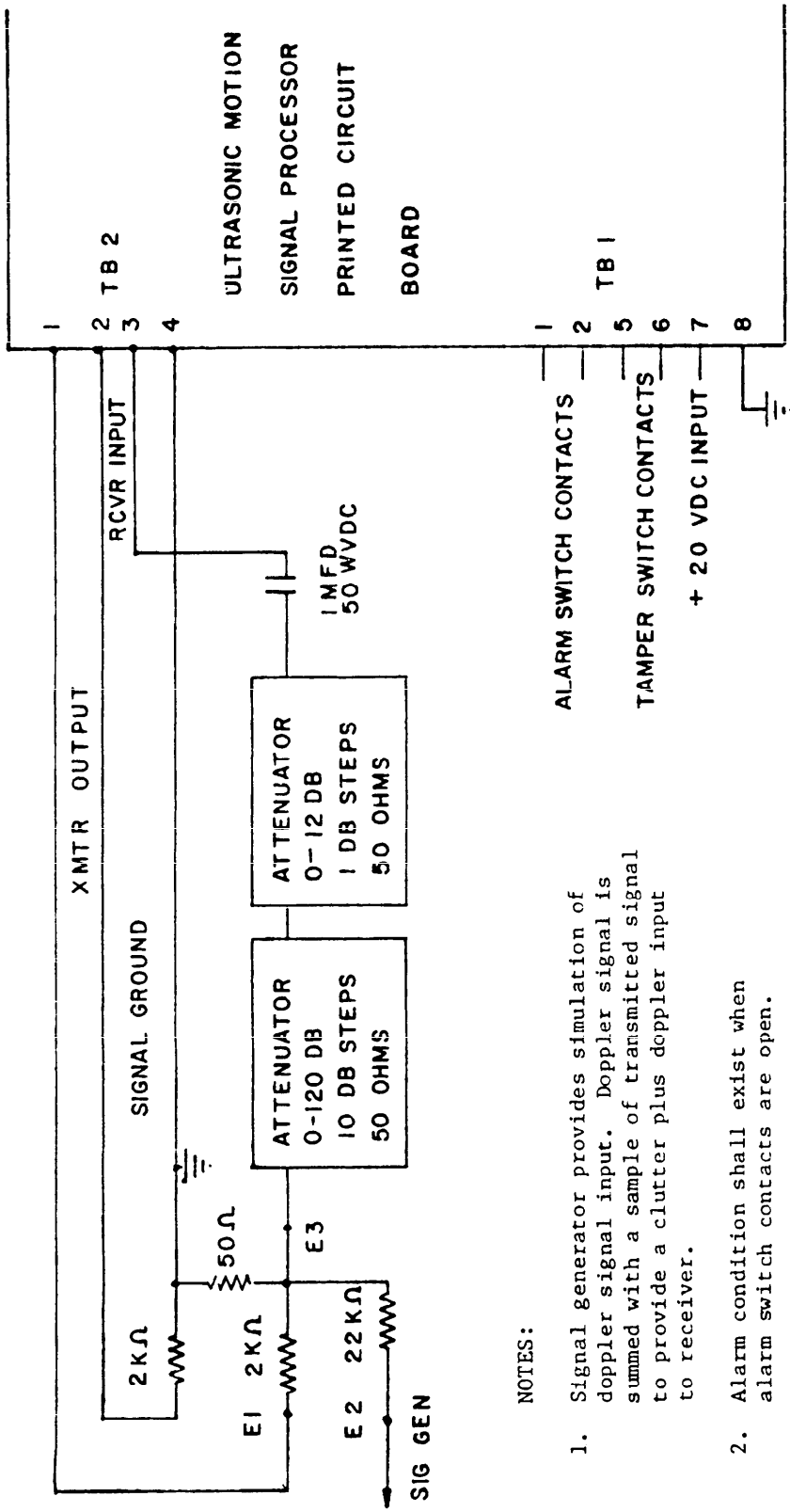
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X-3927A

FIGURE 1. Test fixture for transceiver circuit card assembly.

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NOTES:

1. Signal generator provides simulation of doppler signal input. Doppler signal is summed with a sample of transmitted signal to provide a clutter plus doppler input to receiver.
2. Alarm condition shall exist when alarm switch contacts are open.
3. Tamper condition shall exist when tamper switch contacts are open.

FIGURE 2. Test fixture for ultrasonic motion signal processor printed circuit board.

X-3928

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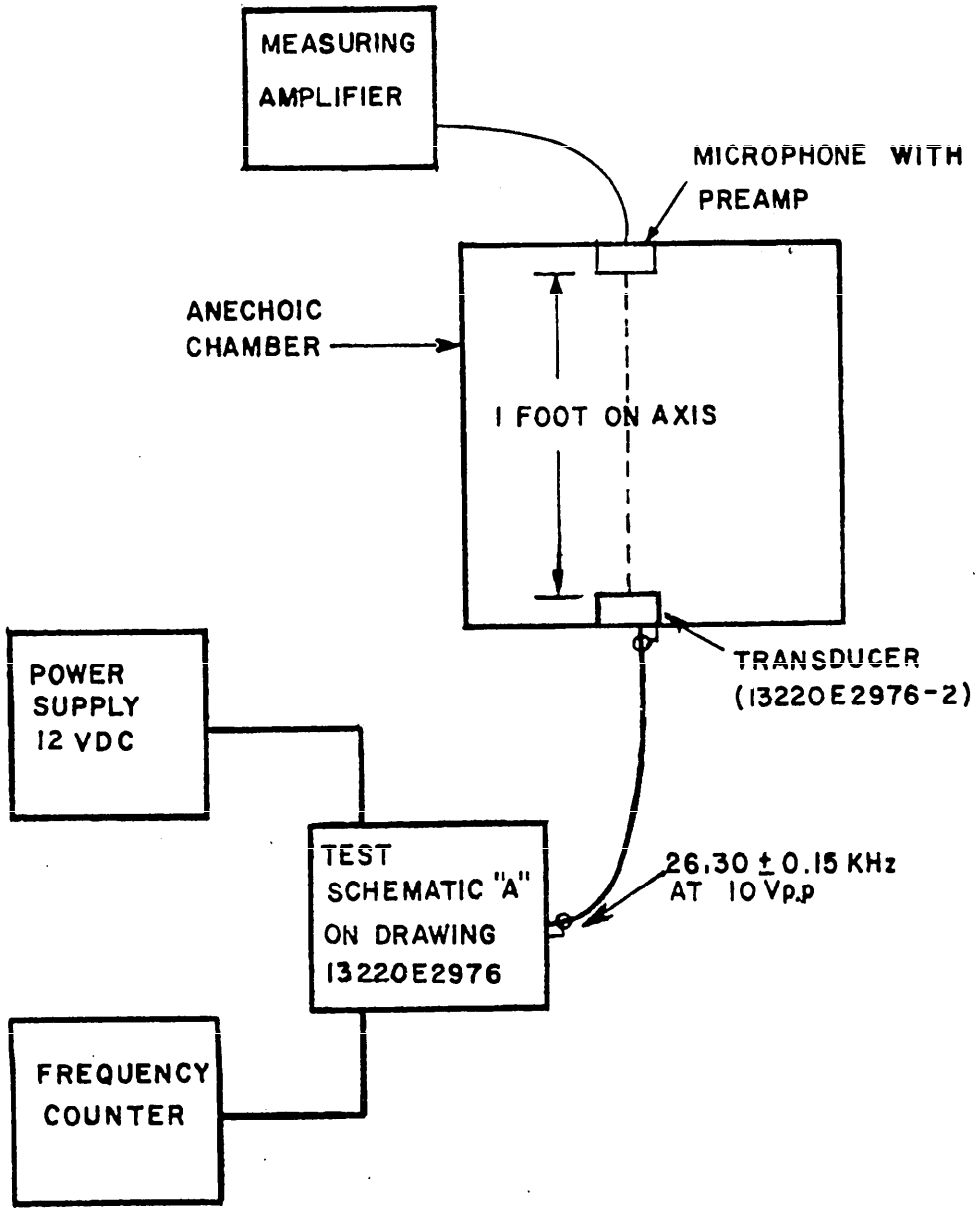


FIGURE 3. Functional test for transducer, (13220E2976-2) red dot.

X-3929

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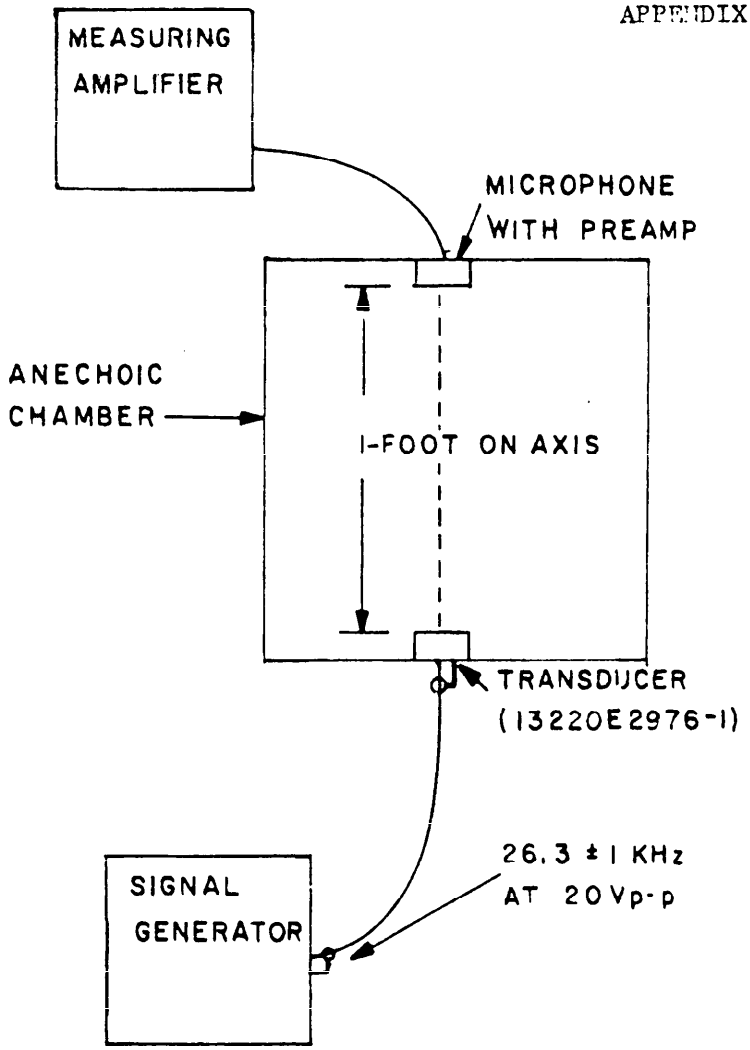


FIGURE 4. Functional test for transducer (13220E2976-1) white dot.

X-3930

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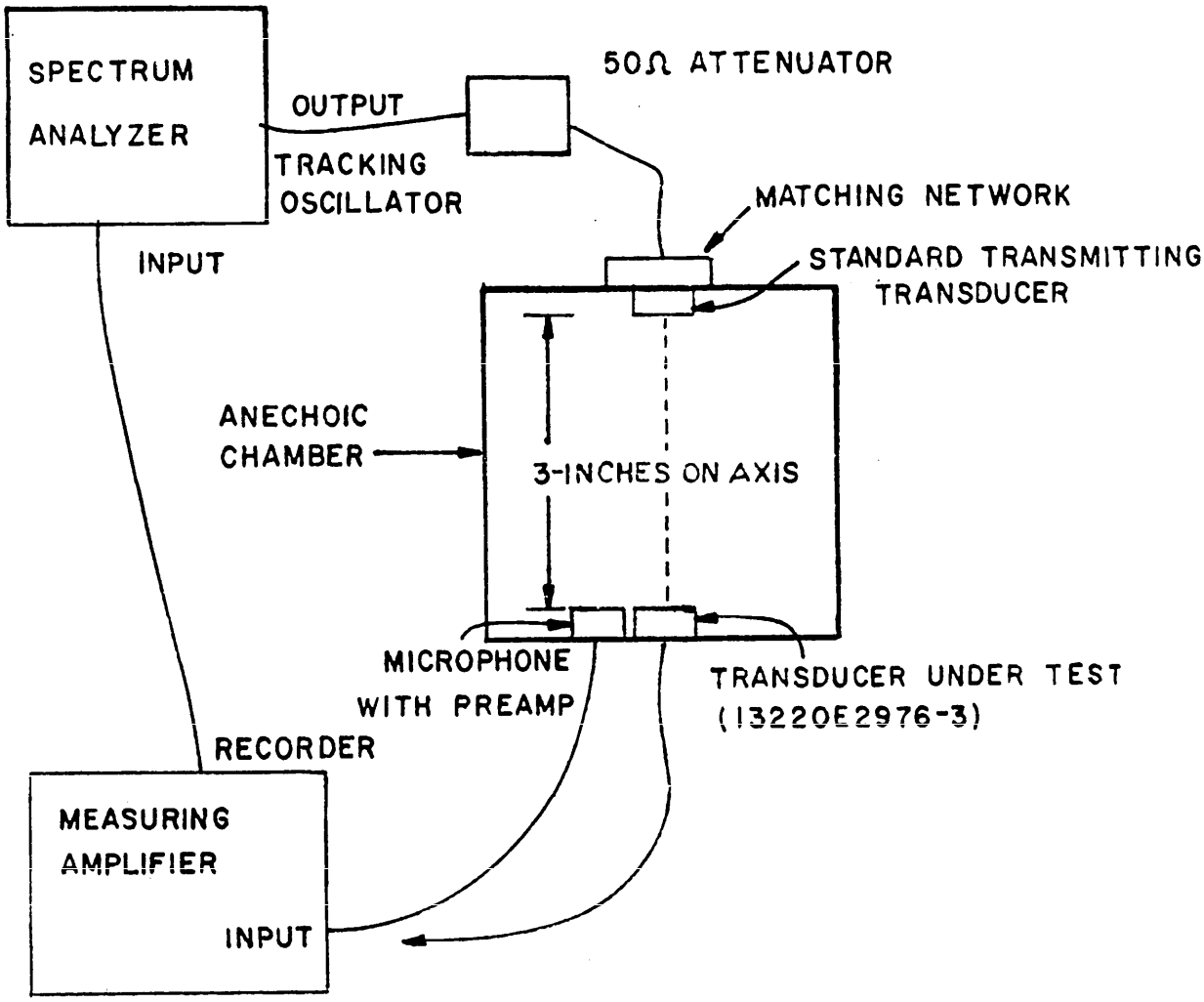
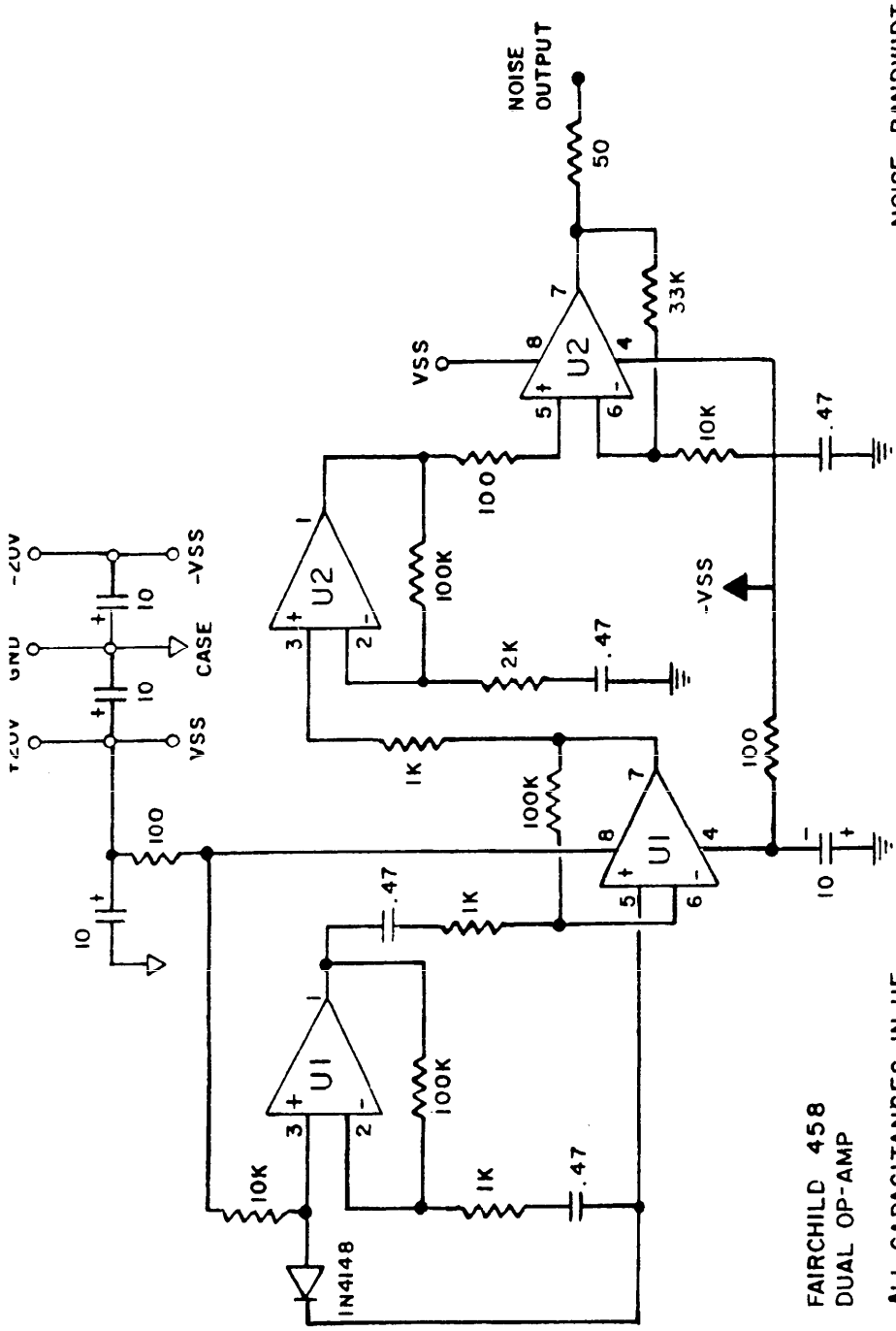


FIGURE 5. Functional test for transducer, (13220E2976-3) blue dot.

X-3931

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NOISE BANDWIDTH
-3dB @ 1-30 KHZ

X-3494A

U1, U2 FAIRCHILD 458
DUAL OP-AMP

ALL CAPACITANCES IN UF

ALL RESISTANCES IN OHMS

FIGURE 6. Ultrasonic noise generator.

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DOCUMENT IDENTIFIER AND TITLE MIL-S-52872B Sensor Group, Ultrasonic Motion: Transceiver RT-1161/FSS-9(V) and Processor MX-9444/FSS-9(V)

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